

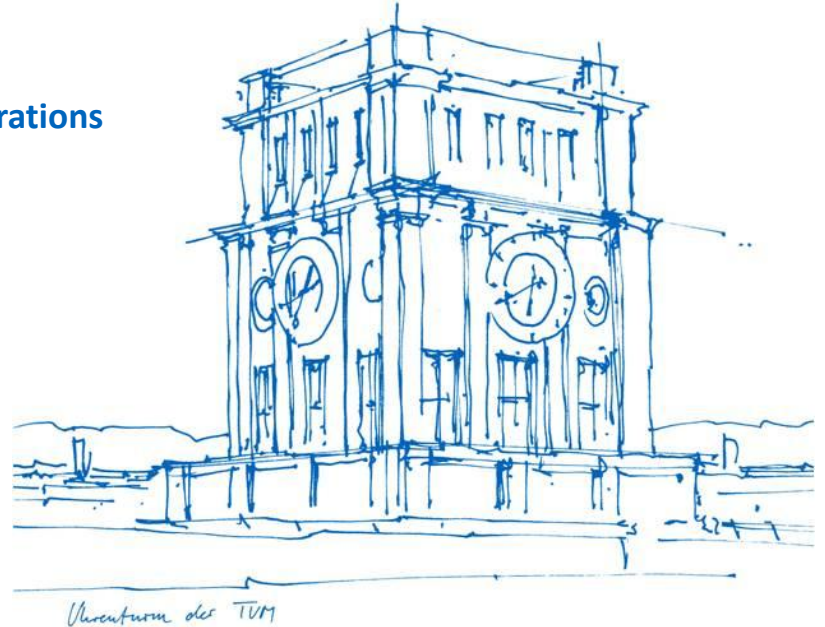
The SAP UCC Munich as an ERP Data-Center Use Case

**Advanced Practical Course – Next Generation Data Center Operations
for SAP Enterprise Software (IN2128, IN2106, IN212810)**

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SAP University Competence Center
Information Systems and Business Process Management
TUM School of Computation, Information and Technology
Technical University of Munich

22 April, 2026



Motivation

Why do we present our UCC use case?

- a. So that you see, how the abstract concepts we spoke about the last three days, can manifest in the real world
- b. Because we know it very well
- c. Because some of your projects will place therein

*“In theory, theory and practice are the same.
In practice, they are not.” – Albert Einstein*

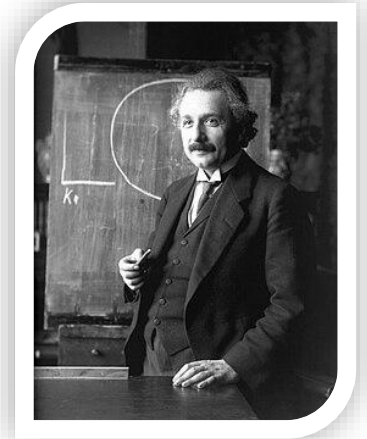


Image Source:
https://de.wikipedia.org/wiki/Albert_Einstein#/media/Datei:Albert_Einstein_1921_by_F_Schmutzer.jpg

Agenda

Preface – The UCC at the TUM

Part I - Company Presentation

The UCC Business Case

The UCC's Business Environment

Part II – The UCC Data Center Setup

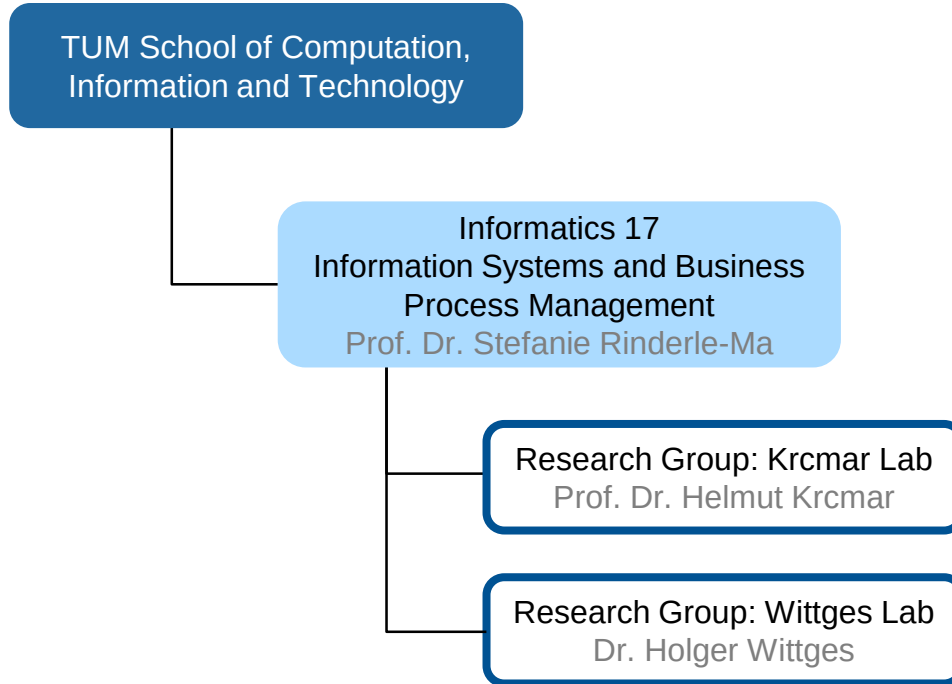
Hardware & Landscape Setup

Network Setup

Backup

Preface – the UCC at the TUM

Organizational Chart



- Chairholder Prof. Rinderle-Ma
 - Process Automation
 - Process Intelligence
 - Compliance Management
- Two independent research groups within the chair
 - Krcmar Lab
 - Wittges Lab
- SAP UCC TU Munich as a faculty project
 - 11 team members work actively on a dissertation topic

Part I – Company Presentation

- I. The UCC Business Case
- II. The UCC's Business Environment



The UCC Business Case

SAP, SAP UA, ACCs, UCCs



SAP SE



SAP University Alliances

Academic Program of SAP SE

Support for practical educational projects – Teaching at educational institutions – Worldwide

> 2500 Educational Institutions

> 500.000 Students

Academic Competence Centers

- Application Support

University Competence Centers

- Hosting
- Development of the teaching / learning environment
- Support

Services

Hosting

Operations, System
administration, Hardware

Teaching Materials

Slide sets, Exercises, Case
studies with model company data
sets, instructor notes

Support

Service Desk, Hotline & E-Mail,
Train-the-Trainer-Courses



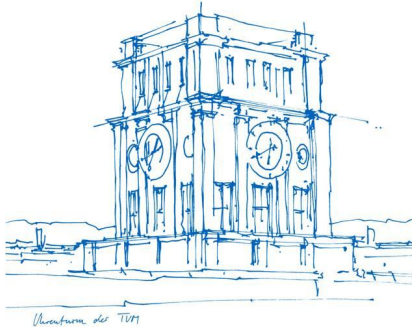
Kooperationen und Forschung

Konferenzen,
Curriculumsentwicklung,
Forschungsprojekte

Key Figures

Launch

October 2, 2003



Dr. Holger Wittges

Managing Director

15

Employees



71.000+

Students

Partners from
20 Countries

2.762

Lecturers



1.000+

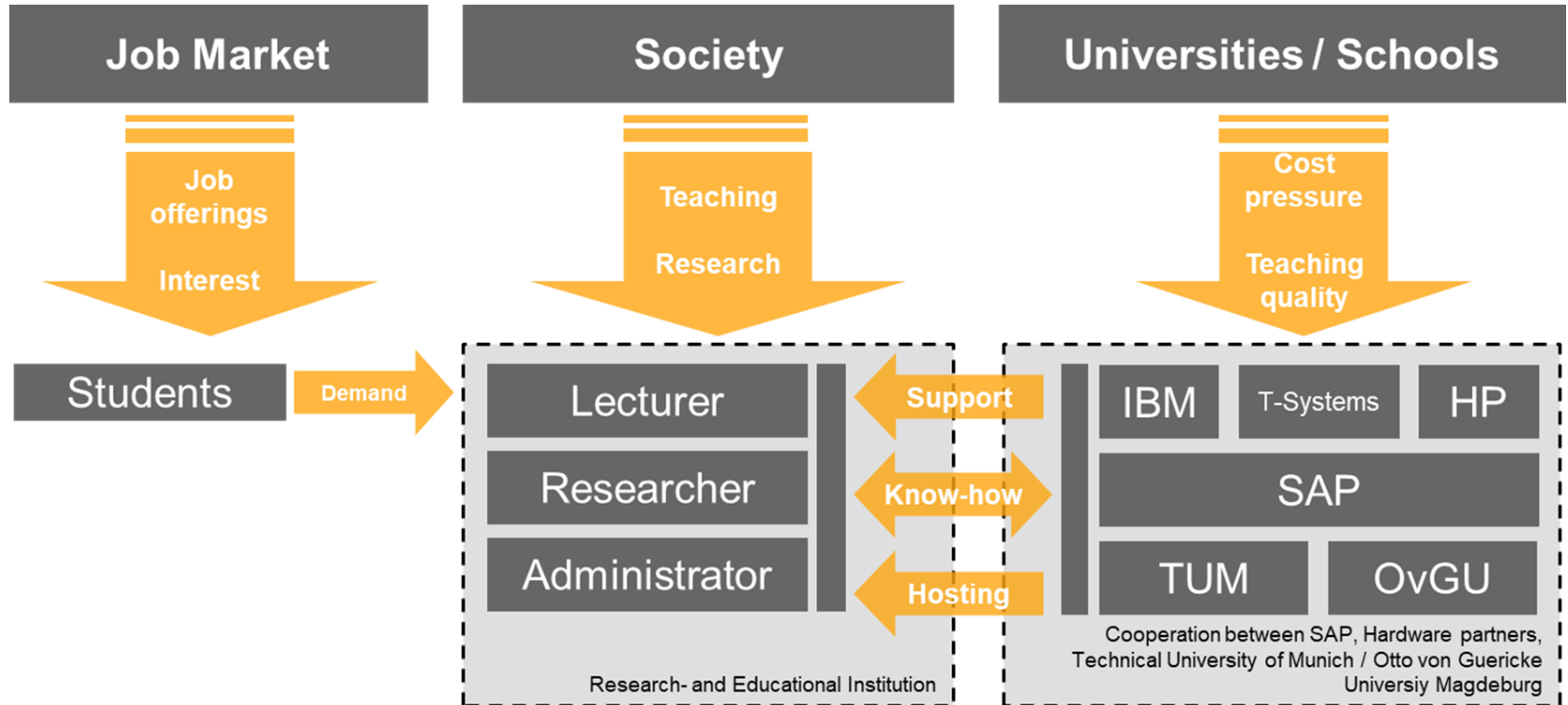
Learning
Environments

516

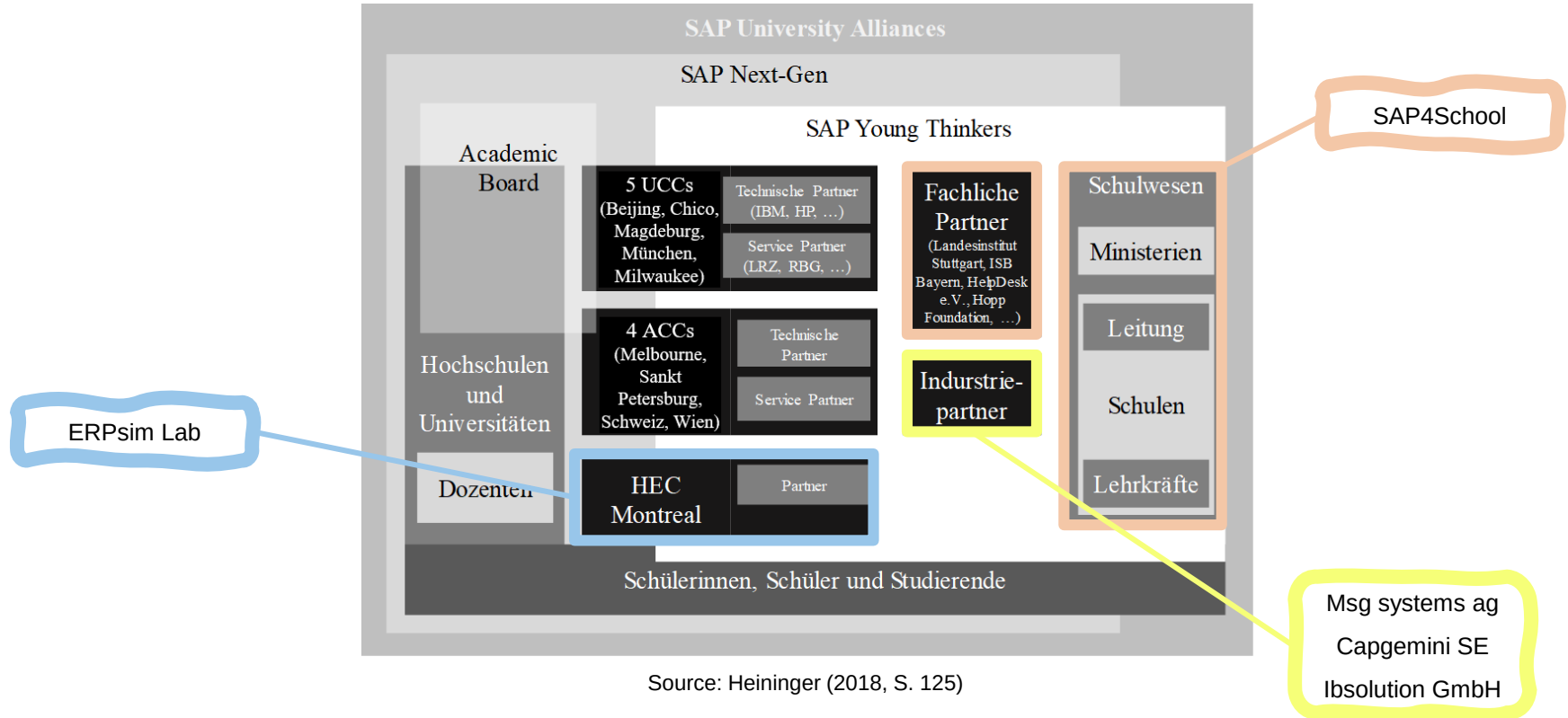
Institutions
are members

The UCC's Business Environment

Ecosystem



Cooperation Partners



Cooperation with SAP – Expectations and Vision

- Long-standing partnership (~25 years) in providing enterprise-grade IT business services
- Joint mission: bringing cutting-edge SAP solutions into teaching and research
- Expectation: continued collaboration as hybrid cloud services grow in relevance

Goal: Strengthen the bridge between academia and industry by making SAP's evolving business services accessible for education and innovation.

Regular Strategic Dialogues

- Ongoing discussions on short-, medium-, and long-term goals
- Shared understanding for future development of the *SAP University Alliances Program*



Part II – The UCC Data Center Setup

I. Hardware & Landscape Setup

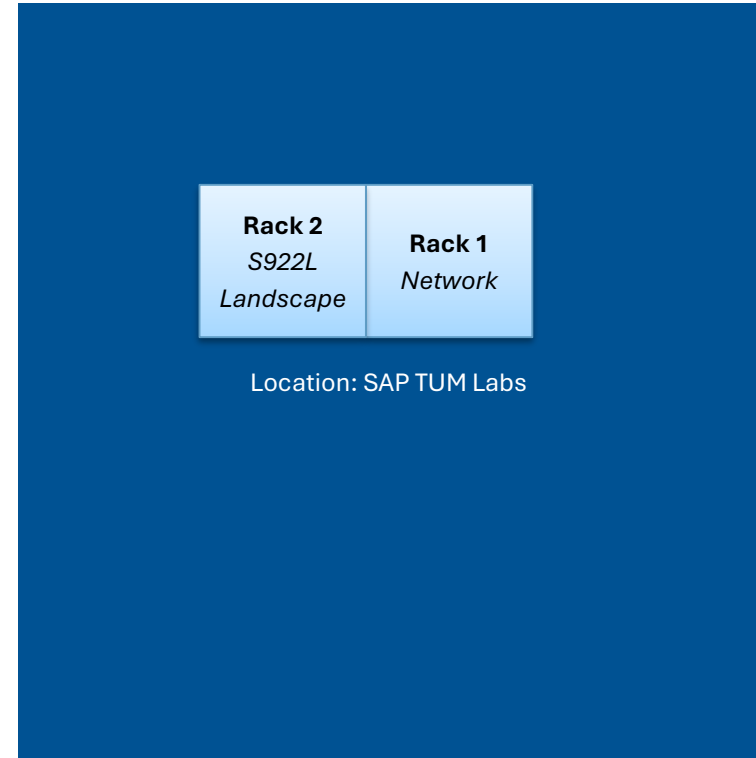
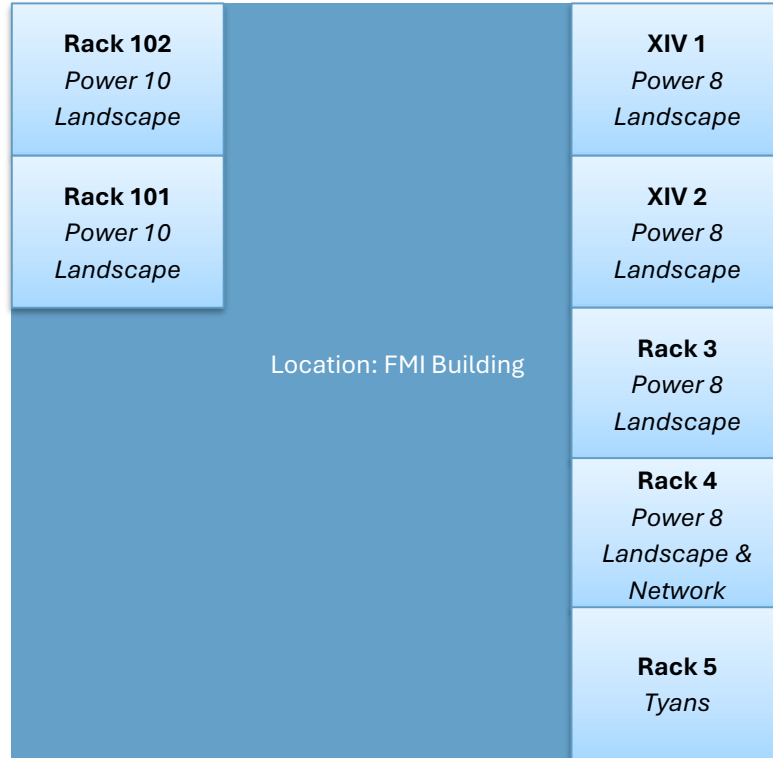
II. Network Setup

III. Backup

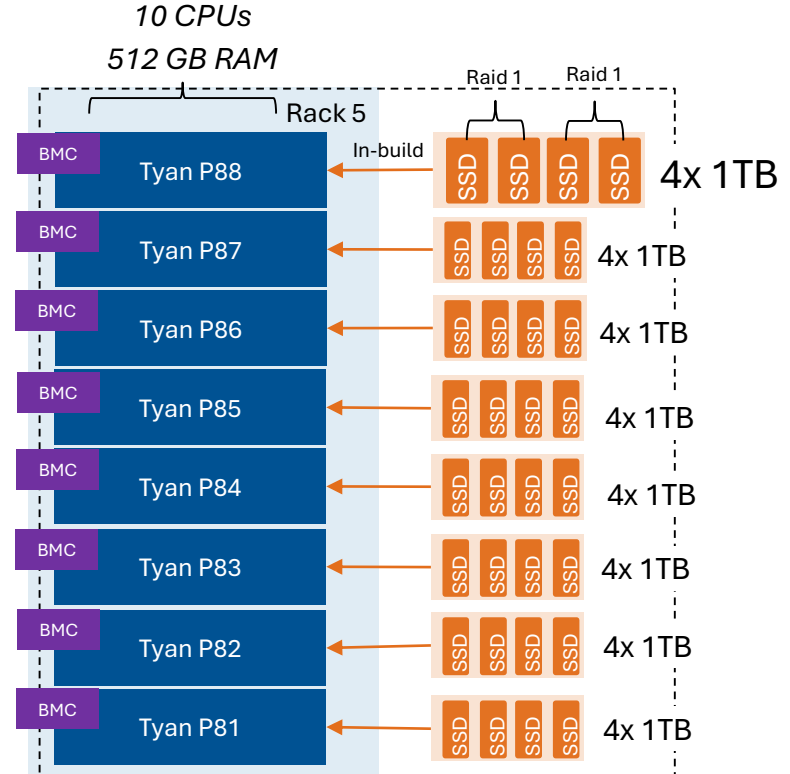
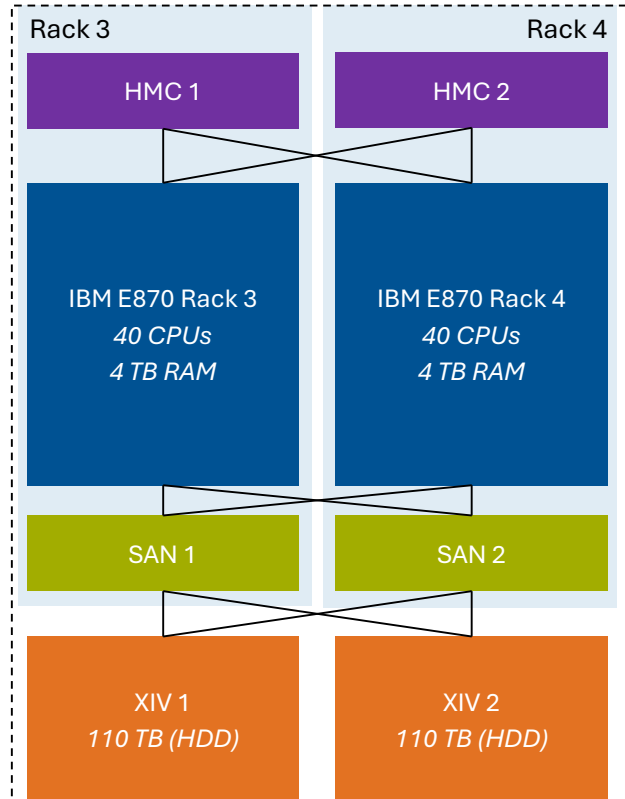


The UCC Hardware & Landscape Setup

UCC Hardware: Data Centers

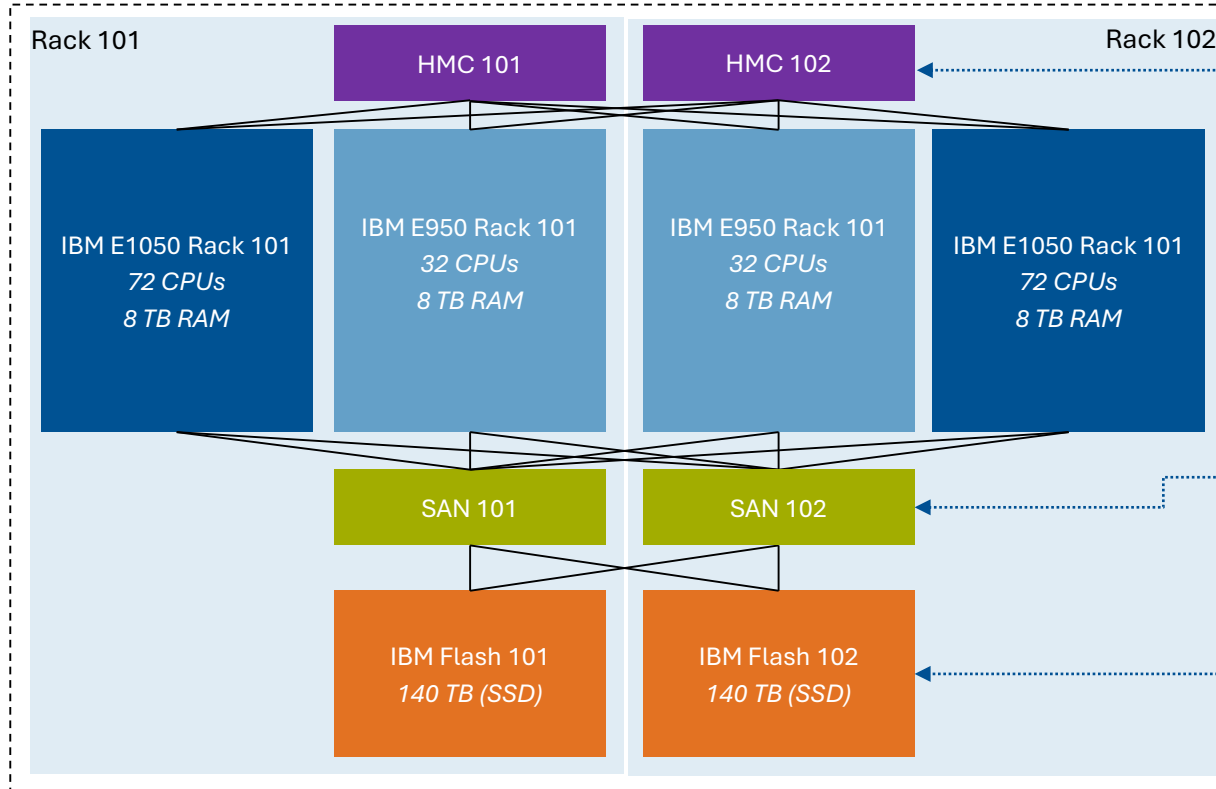


UCC Hardware: Power 8 Landscape



UCC Hardware: Power 10 Landscape

managed automatically

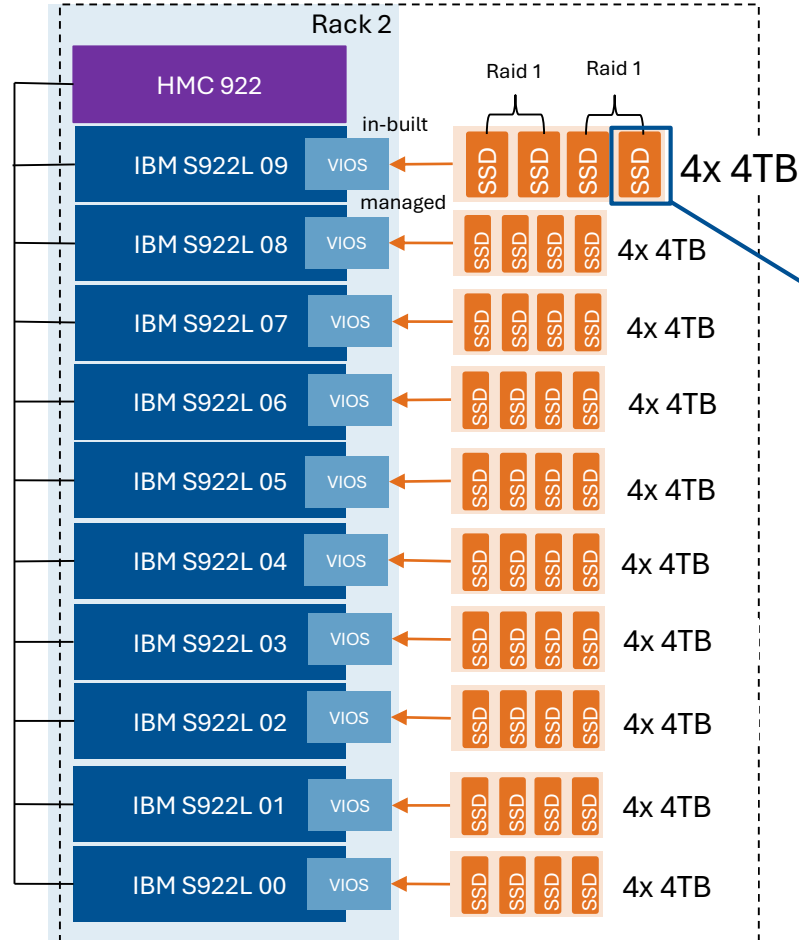


PowerVC

PowerVC logo taken from <https://www.linkedin.com/pulse/creating-shared-volumes-powervc-using-ansible-modules-klyachkin/>

S922L Landscape

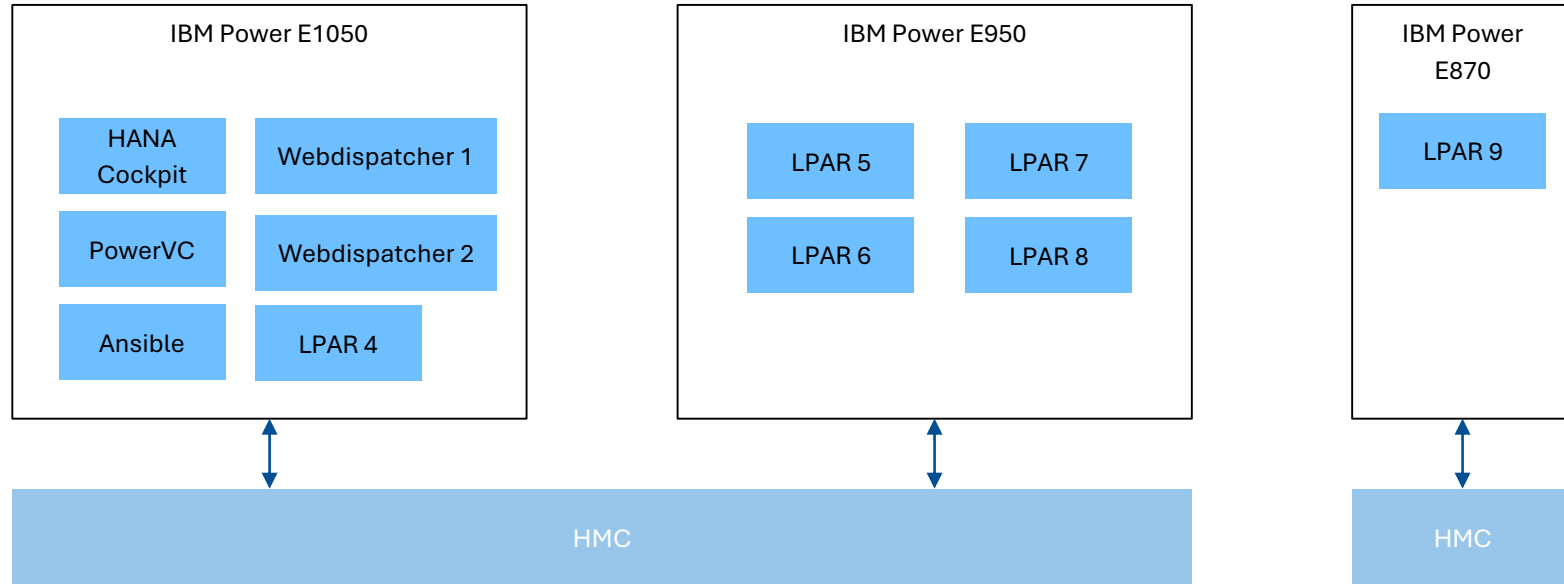
Each S922L:
20 CPUs
512 GB RAM



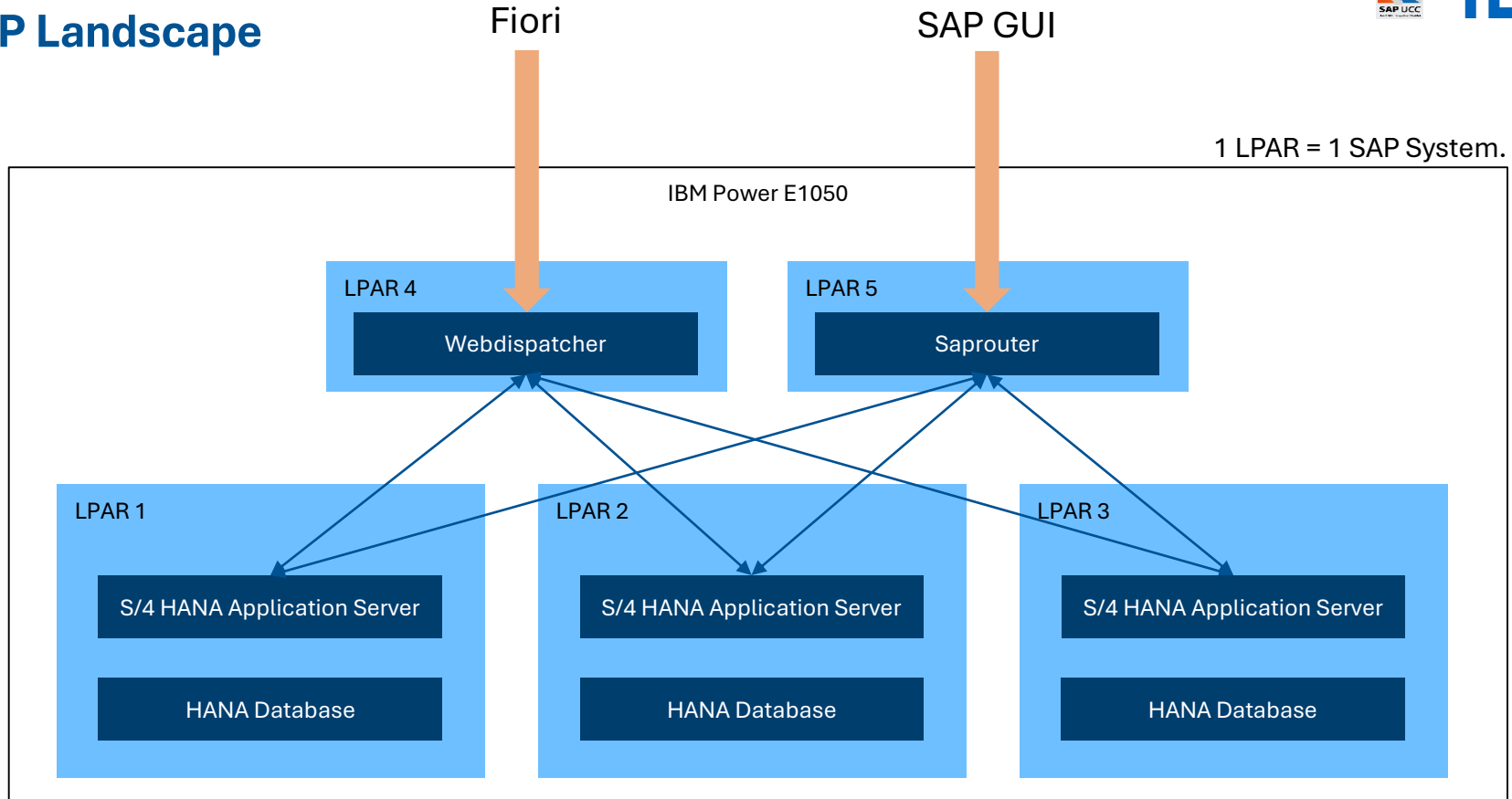
Currently Going Live

LPAR Landscape

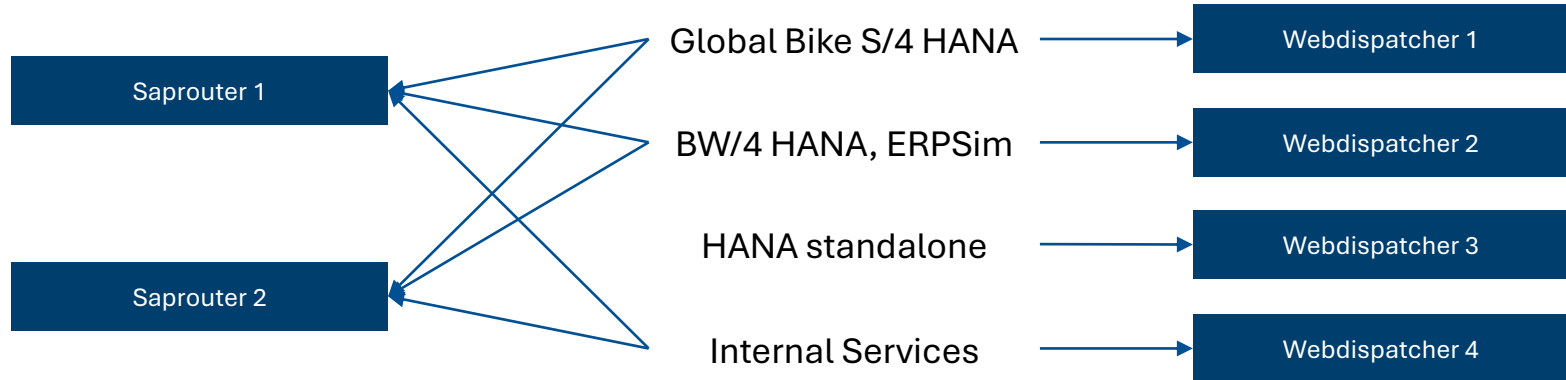
All LPARs have access to all LPARs.



SAP Landscape



External Connection



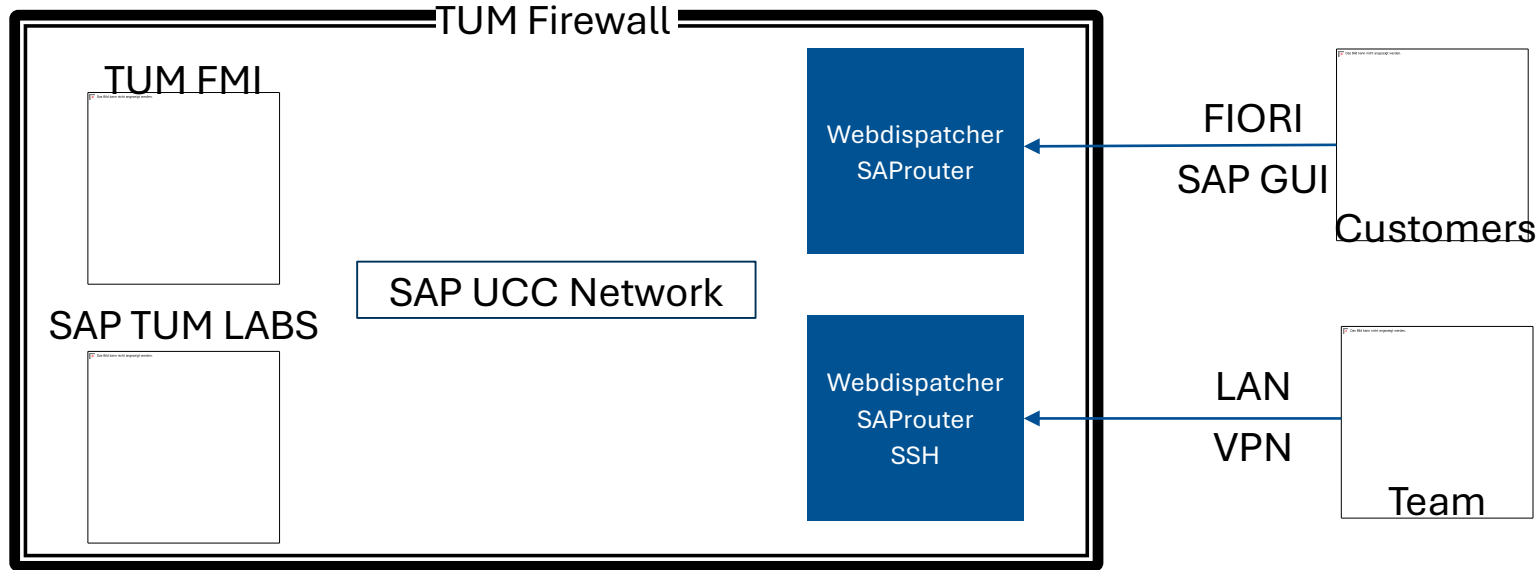
Inside an IBM E1050



Source: <https://www.ibm.com/de-de/products/power-e1050>

Network Setup

Network and Connectivity



Managed by TUM IT Operations (ITO):

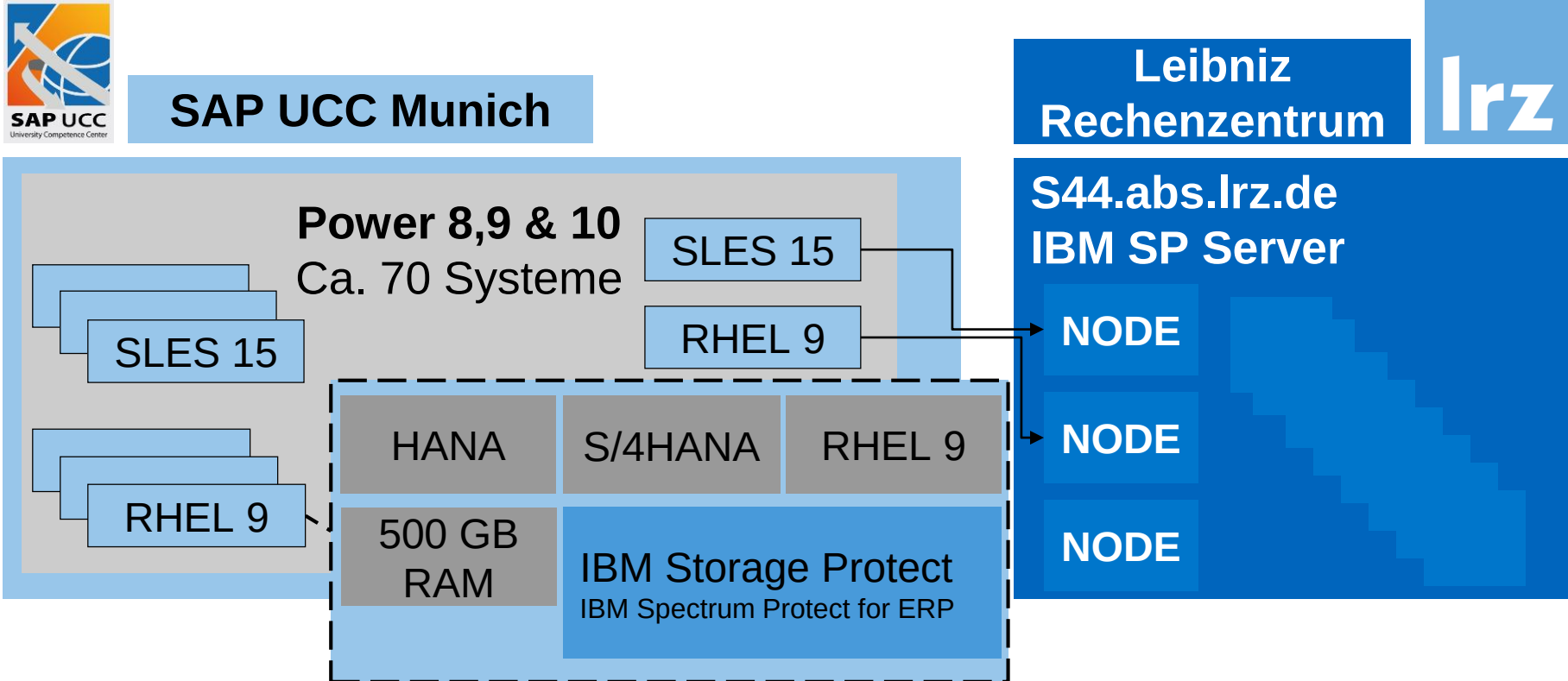
- Network Firewall
- DNS
- Vulnerability Scans

Internet Connection

- The Internet is provided by TUM CIT's ITO group
- Each server room has a dark fiber internet connection
- Each server has 1-4 SFP Modules with a 40 Gbit/s Fibre-Channel Internet Connection each
- Each VIOS has two Ethernet cards merged to a shared Ethernet adapter connected to a virtual switch
 - The Server Room in SAP TUM Labs uses the LACP protocol
 - The Server ROOM in the FMI Building uses the Round-Robin Protocol
- The LPARs have one virtual Ethernet card connected to the virtual switch

Backup

System Landscape



Backup Quality Assurance



Secured Data Center:

- Climate Control
- Fire Protection
- Access Control



Metadata Protection

- Raid 6 / Raid DP
- Daily Backups



High-Quality Components

- Trusted Manufacturers
- Reliable Storage Media



System Expansion as needed

- Server capacity, Storage, Network



Redundant Systems

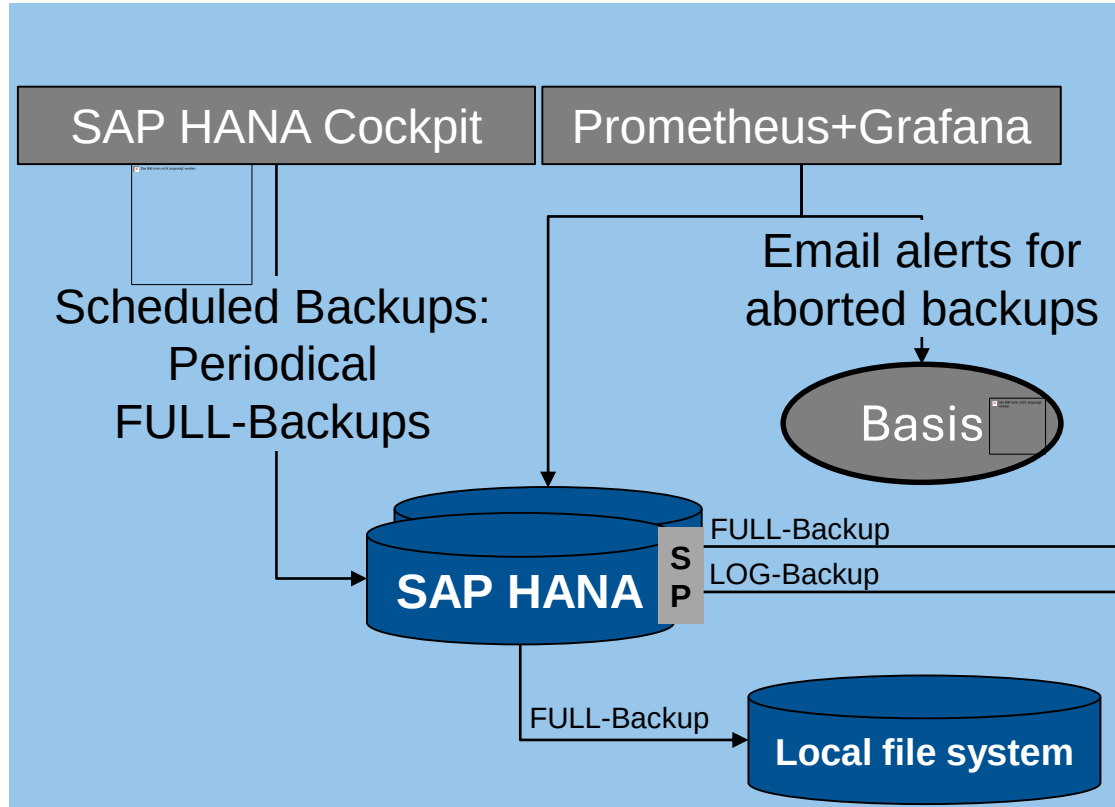


Continuous Upgrades

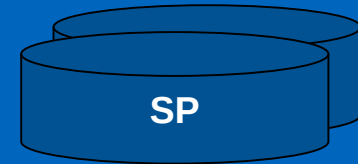


Regular Tape Migration

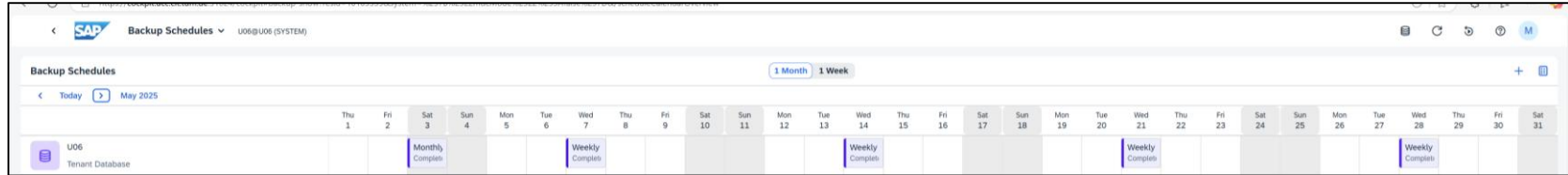
Backup



LRZ
IBM SP Server



Backup Strategy



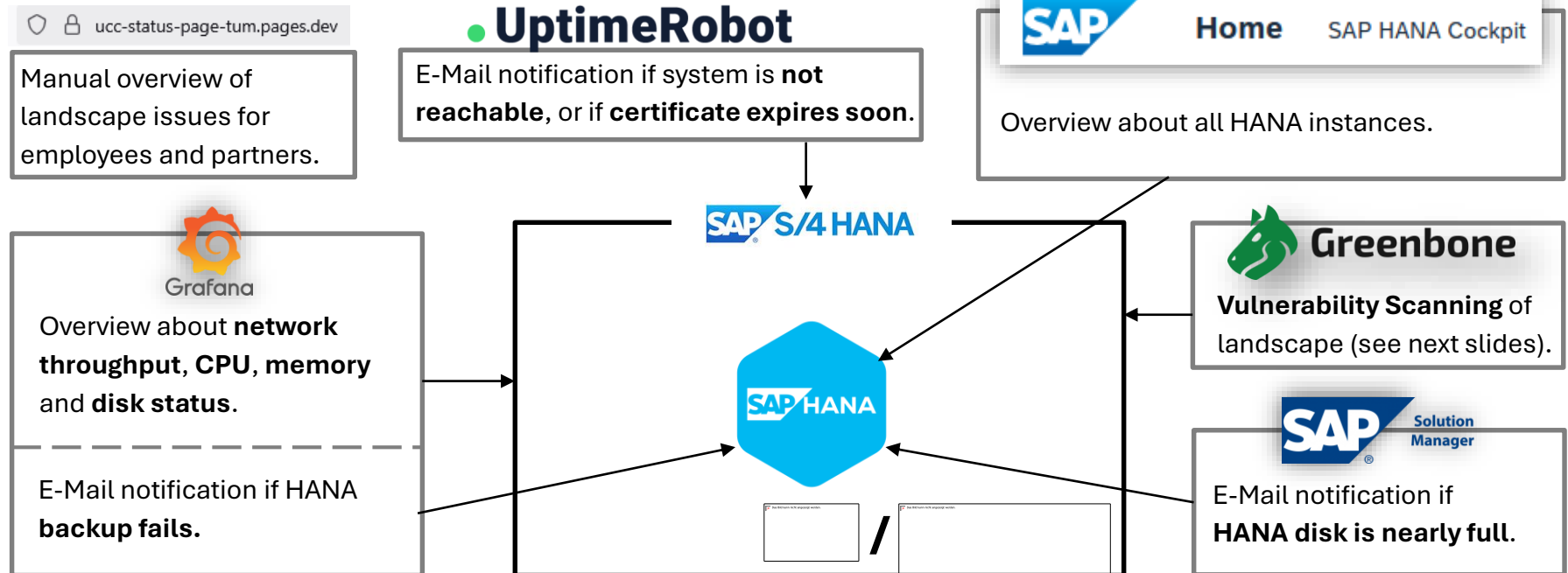
Weekly: complete data backup in backint
Monthly: complete data backup in local file system

Recovery Strategy

- Documented Procedures:
 - Emergency Shutdown/Reboot, Importing Backups, System Replication from Backup
- Documented Crisis Roles:
 - Power of Decision, Incident Investigation, Communication, Support

Incident	(Recovery) Strategy	RTO
Data loss – Inconsistency	Import Backup	~ 2-3 Hours
LPAR loss – OS damage	System Replication from Backup	~ 1-2 Days
Servers offline – TUM FMI Data Center offline	Replacement Systems in SAP TUM LABS (crucial systems only)	~ 2-3 Hours

Technical Quality Assurance (Monitoring)



[1][2][3][4][5][6][7][8]

Server Room Visit

Thank you for your attention!

